Tutorial 3 Answers, Embedded Systems

① a) start 1 1 0 0 1 0 1 0 stop stop.

LSB MSB

→ 0b 0 1 0 1 0 0 1 1

= 64 + 16 + 2 + 1 = 83 → 'S'

b) Baud rate = 9600 bits/s.

8 data bits, 1 start bit, 2 stop bits.

= 11 bits / integer.

So, $\frac{\text{integers}}{\text{second}} = \frac{9600 \text{ bits/s}}{11 \text{ bits/integer}}$

= 872 integers/s.

(2)

a) Lines 26 - 27

b) Add lines:

```
button.o : button.c button.h
    ⟶ $(CC) -c $(CFLAGS) $< -o $@
```

Modify line 26 to:

```
flash.out : flash.o pio.o system.o led.o button.o
```

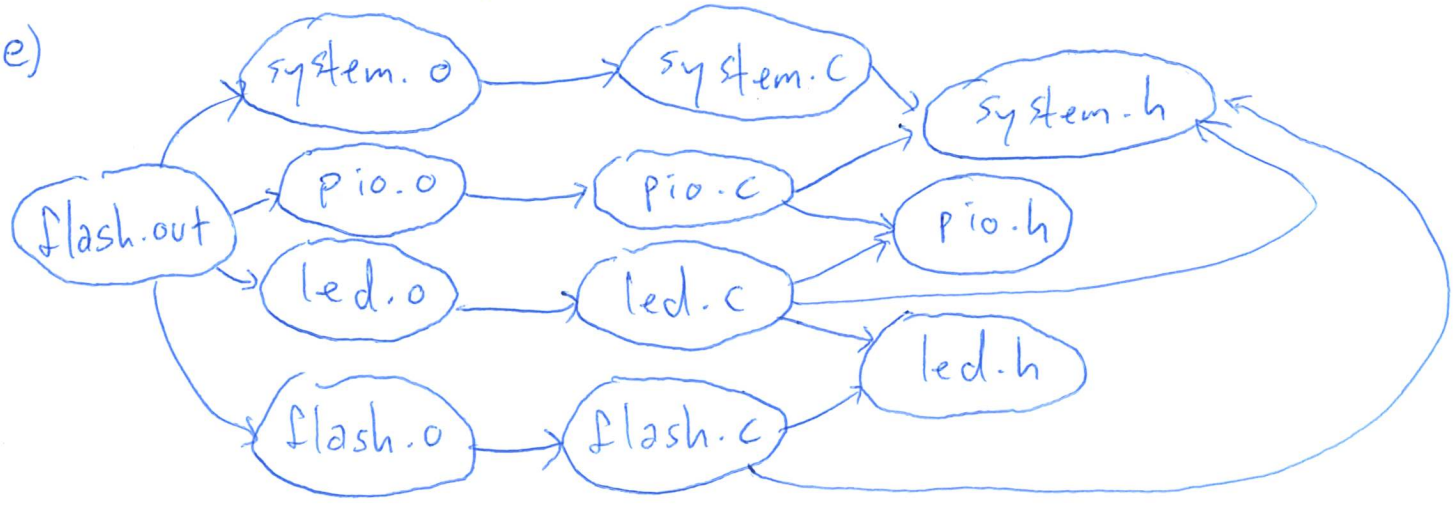
c) These lines declare a phony target, which isn't the name of a real file. However, the target can still be "made" (here using 'make clean'), which will run the command

'rm *.o *.out *.hex'.

This will delete any present object files (.o) or executable binaries (.out, .hex), and so clean all build products from the directory.

d) OBJCOPY here refers to 'avr-objcopy', which converts between executable binary formats. Here we are using it to convert a .out file to the .hex format, which is what the ATmega32U2 uses.

e)



(3)

a) C source files: bounce1.c, system.c, timer.c, pacer.c, pio.c.

Header files: pio.h, system.h, pacer.h, timer.h

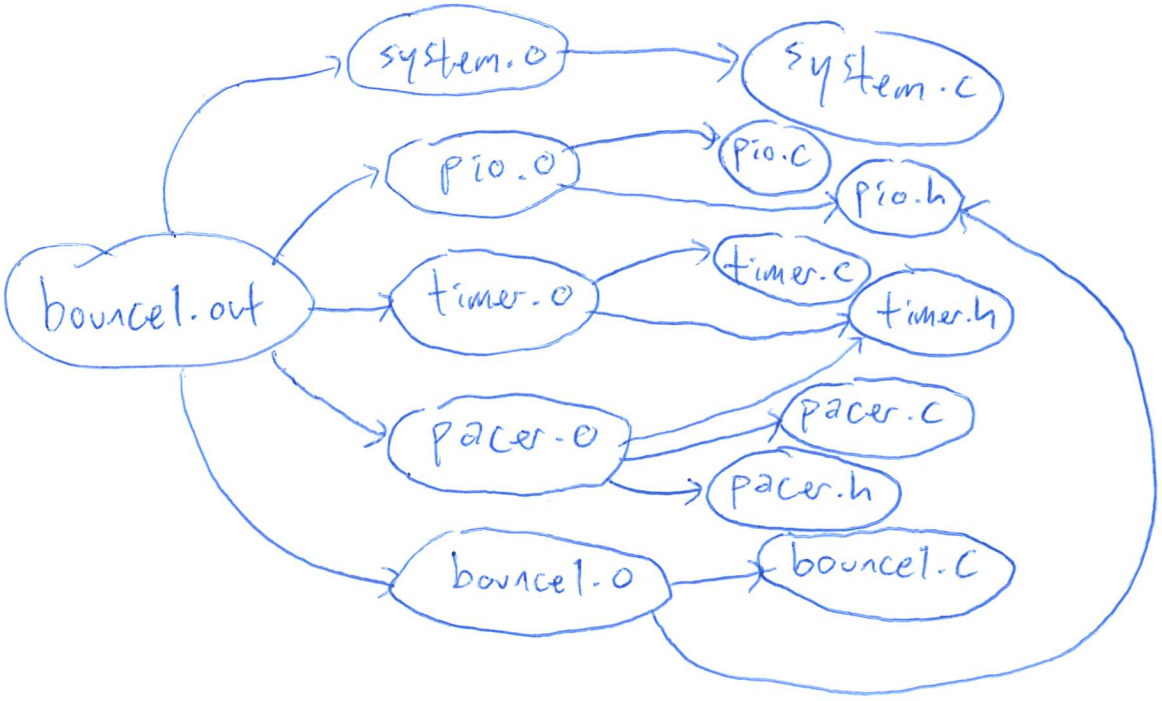
object files: bounce1.o, system.o, timer.o, pacer.o, pio.o.

Executable: bounce1.out.

b) C source files & header files. Not object or executable files.

c) 'make all' will compile all source files into respective object files, & link the object files into the executable bounce1.out.

d)



e) avr-gcc -mmcu=atmega32u2 -Os -Wall bounce1.o pio.o system.o timer.o pacer.o -o bounce1.out -lm

(4)

a) 'continue' indicates a deliberately empty loop in C, which ensures that the compiler does not optimize away the while loop.

b)

```
bool toggler=0;
while(1)
{
    pacer_wait();

    if (toggler)
    {
        pio_output_high(cols[2]);
        pio_output_high(rows[3]);
        pio_output_low(cols[1]);
        pio_output_low(cols[3]);
        pio_output_low(rows[2]);
        pio_output_low(rows[4]);
    }
    else
    {
        pio_output_high(cols[1]);
        pio_output_high(cols[3]);
        pio_output_high(rows[2]);
        pio_output_high(rows[4]);
        pio_output_low(cols[2]);
        pio_output_low(rows[3]);
    }
    toggler= !toggler;
}
```

c) git diff

d) git restore .